

TASK 3: Automation System Design

Smart Home Lighting System — Flowchart & Explanation Report

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1. Introduction

This report presents the design of an automated Smart Home Lighting System. The system uses ambient light sensing and motion detection to automatically control indoor lighting, improving energy efficiency, convenience, and security. The design follows a simple closed-loop control architecture suitable for implementation with low-cost microcontrollers such as Arduino or ESP32 and can be extended with IoT connectivity for remote monitoring and control.

2. System Objectives

- Automatically turn lights ON when the room is dark and motion (occupancy) is detected.
- Automatically turn lights OFF after a configurable period of inactivity or when sufficient ambient light is present.
- Minimize unnecessary energy consumption while maintaining user comfort and safety.
- Provide a modular design that can be expanded to multiple rooms or integrated with voice assistants (Alexa, Google Home).

3. Hardware Components

Component	Function
LDR (Light Dependent Resistor)	Measures ambient light intensity
PIR Motion Sensor	Detects human presence / movement
Microcontroller (Arduino Uno / ESP32)	Processes sensor data and executes control logic
Relay Module (5V)	Switches AC power to lights safely
LED / Smart Bulb / Room Lights	Actuator – illumination
Power Supply (5V / 12V)	Powers the control circuit
Optional: Wi-Fi Module / MQTT	Enables remote control and status monitoring

4. Working Principle & Control Logic

The system continuously monitors two primary inputs: ambient light level (via LDR) and motion (via PIR sensor). The control algorithm is as follows:

1. Read current ambient light value from the LDR.
2. Compare the light value against a predefined threshold (calibrated for the specific room).
3. If the room is sufficiently bright → keep lights OFF (or turn them OFF if previously ON).
4. If the room is dark → check the PIR sensor for motion.
5. If motion is detected → activate the relay to turn the lights ON and start a countdown timer (e.g., 3–10 minutes).

6. While lights are ON, any new motion detection resets the timer.

7. When the timer expires without further motion → turn the lights OFF and return to continuous monitoring.

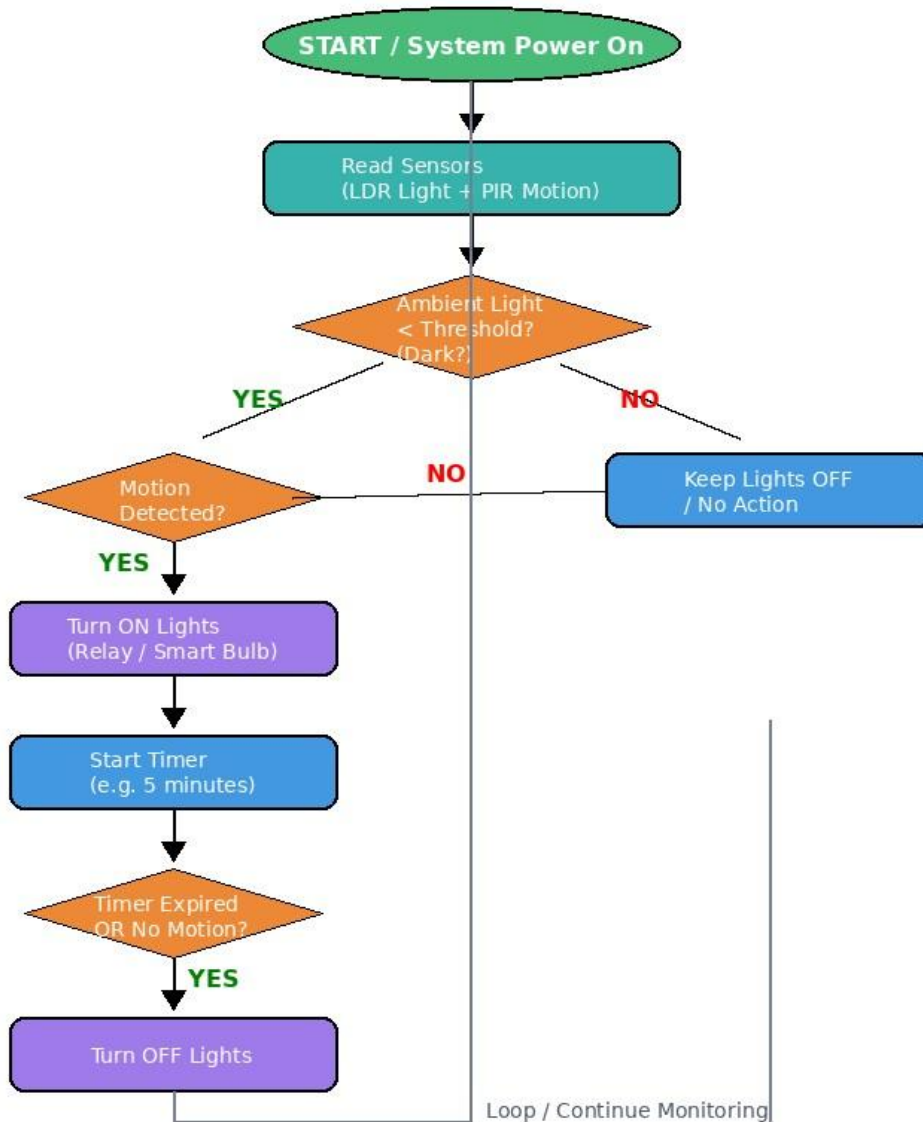
This logic ensures lights are used only when needed, combining both environmental and occupancy conditions.

5. System Flowchart

The complete decision flowchart of the automation system is shown below. It illustrates the start-up sequence, sensor reading loop, decision diamonds for light and motion, actuator actions, and the feedback loop that continuously monitors the environment.

Smart Home Lighting Automation System

Flowchart / Logic Diagram



Legend:



Components: LDR (Light Dependent Resistor), PIR Motion Sensor, Microcontroller (Arduino/ESP32), Relay |
Optional: IoT (WiFi/MQTT) for remote control via smartphone app / Google Home / Alexa

Figure 1: Flowchart of the Smart Home Lighting Automation System

6. Basic Circuit / Connection Overview

- LDR is connected to an analog input pin of the microcontroller through a voltage-divider circuit (typically 10 kΩ resistor).
- PIR sensor digital output is connected to a digital input pin (with optional pull-down).

- Microcontroller digital output pin drives the relay module (optically isolated recommended for AC loads).
- Relay switches the live AC line to the lighting load. Neutral remains common. Proper isolation and safety precautions must be observed when working with mains voltage.
- For IoT extension: ESP32 can publish sensor status and receive commands via MQTT broker (e.g., Mosquitto) or integrate directly with Home Assistant / Blynk / Arduino IoT Cloud.

7. Advantages of the Proposed System

- Significant energy savings by eliminating lights left on in unoccupied or well-lit rooms.
- Improved convenience — no need to manually switch lights when entering a dark room.
- Enhanced security — lights activate automatically if motion is detected at night.
- Low cost and easy to implement with widely available components.
- Scalable: can be extended to multiple zones, dimming control, or integration with other smart-home devices.

8. Limitations and Possible Improvements

Current limitations include sensitivity to ambient light changes (e.g., sudden cloud cover) and possible false triggers from pets. Future enhancements may include:

- Adaptive threshold using machine learning or time-of-day schedules.
- Multi-sensor fusion (ultrasonic + PIR) for better occupancy detection.
- Integration with voice assistants and mobile app for override and scheduling.
- Energy monitoring and logging for analytics.

9. Conclusion

The Smart Home Lighting Automation System demonstrates a practical application of sensors, microcontrollers, and simple decision logic to create an energy-efficient, user-friendly lighting solution. The accompanying flowchart clearly defines the control sequence, making the design easy to implement, debug, and expand. This project aligns with the goals of modern home automation and Industry 4.0 principles of intelligent, connected systems.

— End of Report —

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